Technical Report

1 PROOF OF LEMMA 4.1

Lemma 4.1. Optimizing the loss function in Equation 3 is equivalent to maximizing $I(f_e(\mathcal{G}[S_1]); f_e(\mathcal{G}[S_2]))$, leading to the maximization of $I(f_e(\mathcal{G}[S_1]); \mathcal{G}[S_2])$, where I represents mutual information.

Proof. Since the objective of the encoder is to maximize the embedding similarity between positive subgraph pairs and minimize the embedding similarity between negative pairs, the loss function for a pair of subgraphs $G[S_1]$ and $G[S_2]$ with embeddings $e(G[S_1])$ and $e(G[S_2])$ is equivalent to:

$$\mathcal{L}_{1,2} = -\log \frac{\exp(-D(G[S_1], G[S_2])}{\sum_{i,j} \exp(-D(G[S_i], G[S_j]))} . \tag{1}$$

And minimizing Equation 1 is equivalent to maximizing a lower bound of the mutual information between the learned embeddings of two subgraphs as proven in [5], which is equivalent to maximizing the mutual information between their embeddings $I(f_e(\mathcal{G}[S_1]); f_e(\mathcal{G}[S_2]))$. According to Lemma A.5 in [4], it leads to the maximization of $I(f_e(\mathcal{G}[S_1]); \mathcal{G}[S_2])$. □

2 PROOF OF THEOREM 4.2

Theorem 4.2. Let f_{e_1} denote our proposed abnormal subgraph encoder with financial distribution considered, and let f_{e_2} denote the encoder without financial distribution modeling as in [2, 3]. After sufficient training of f_{e_1} and f_{e_2}, $I(f_{e_1}(\mathcal{G}[S_1]); y) > I(f_{e_2}(\mathcal{G}[S_1]); y)$, where y represents the graph label.

Proof. The encoder f_{e_1} can differentiate financial subgraphs with different financial distributions that f_{e_2} cannot. It means that f_{e_1} can capture more information of subgraphs $\mathcal{G}[S_1]$ than f_{e_2}:

$$H(\mathcal{G}[S_1]) \geq H(f_{e_1}(\mathcal{G}[S_1])) > H(f_{e_2}(\mathcal{G}[S_1])) , \tag{2}$$

where $H(\cdot)$ represents information entropy. Since $f_{e_1}(\mathcal{G}[S_1])$ and $f_{e_2}(\mathcal{G}[S_1])$ are functions of $\mathcal{G}[S_1]$, we have

$$I(f_{e_1}(\mathcal{G}[S_1]); \mathcal{G}[S_1]) > I(f_{e_2}(\mathcal{G}[S_1]); \mathcal{G}[S_1]) . \tag{3}$$

According to Theorem A.6 in [4] and Lemma 4.1, it holds that $I(f_{e_1}(\mathcal{G}[S_1]); y) > I(f_{e_2}(\mathcal{G}[S_1]); y)$. □

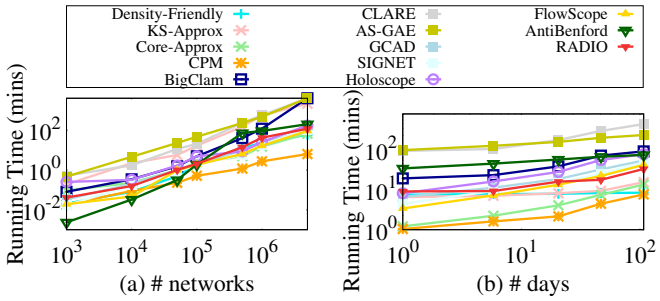


Figure 1: Scalability on (a) # networks; (b) # days; efficiency on (c-d) # abnormal subgraphs on ETH-Jan-18, 19.

3 SCALABILITY EVALUATION

We use Intel Xeon Platinum 8370C processor 32-core 2.8GHz CPU with NVIDIA GeForce RTX 4090 24G GPU for evaluation. We use five datasets spanning from 1 week to 1 year of 2019 to analyze the running time of our method in Section 3.2 as in [1]. Details are in Table 1.

Table 1: Statistics of experimented datasets.

Datasets	# Nodes	# Edges	# Transactions
ETH-1 Week	779,237	1,016,481	1,438,641
ETH-1 Month	2,199,347	3,331,594	6,128,061
ETH-3 Month	2,877,055	8,504,999	17,529,925
ETH-6 Month	5,018,047	21,572,152	43,885,527
ETH-1 Year	6,820,719	38,917,136	85,055,054

3.1 Scalability w.r.t. the size of graph nodes

We evaluate the scalability of RADIO with respect to the size of graph nodes in Figure 1 (a). We find that the running time of RADIO is approximately linear w.r.t. the size of financial networks in the log-scale axes, which means that the time complexity of RADIO is polynomial.

3.2 Scalability w.r.t. the span of time dimension

We also evaluate RADIO's scalability w.r.t. the time dimension. A longer time span results in a larger size of transaction edges $|E|$ when we fix the size of nodes. Figure 1 (b) shows the running time trend as the time span increases from 7 to 365 days. The trend is approximately linear with respect to the time span. The graph node size is fixed at 500,000 to ensure the observations relate only to the time dimension change.

4 SENSITIVITY ANALYSIS OF HYPERPARAMETERS

We omit other datasets due to similar trends to ETH-Jan-18. Similarly, we show $d_a(S)$ results due to similar trends in other metrics.

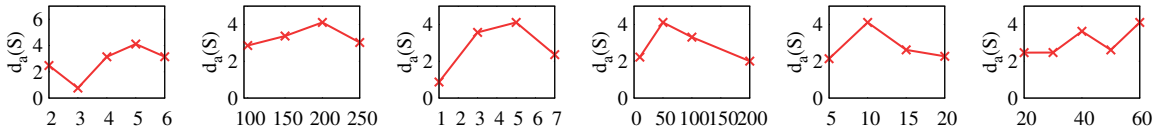
Depth l of Graph Neural Networks. Figure 2 (a) shows that increasing l leads to worse result when $l = 3$, after that achieves the best at 5, then works worse afterward for the over-fitting problems.

Maximum Size m of Nodes in the Boundary. Figure 2 (b) shows that increasing m beyond a certain point, i.e., 200, leads to more noise, resulting in degraded results.

Layer of the k-ego Net. Figure 2 (c) shows that enlarging the layer of the k-ego net beyond a certain point, i.e., 5, brings large-size subgraphs with lower abnormal degrees, leading to poor effect.

Size of the Top-p Candidates. Figure 2 (d) shows that increasing the size of candidates beyond a certain point, i.e., 50, results in more sub-optimal subgraphs, generating worse performance.

Number of Sample Pairs. Figure 2 (e) shows that increasing the number of sample pairs beyond a certain point, specifically 10, results in degraded performance due to potentially increased noise.



(a) Depth l (b) Boundary size m (c) Layer k (d) Size p (e) # sampling pairs (f) # training data

Figure 2: Hyperparameter effects of RADIO on ETH-Jan-18 dataset.

Number of Training subgraphs. Figure 2 (f) shows that enlarging the number of training subgraphs initially improves performance but eventually leads to increased variability.

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